FACET-RHG at Full-Year Scale

Iteration-Free Multi-Agent RTM Clearing over ERCOT 2025 (362 days)

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Abstract

The two 7-day runs in the main report are extended to the **whole of 2025**: the 6-agent green-hydrogen real-time market is cleared by online FACET on **362 days** (3 DST/data-gap days excluded), 34,752 **dispatch steps**, and benchmarked step-for-step against a tuned ADMM ($\rho = 0.01$) on the *identical* parameter sequence, warm-start block. Two conclusions. First, the *communication* claim is robust across every month: FACET clears a step in a **median of one** inter-agent round versus ADMM's ∼**67**, winter to summer, spending 1.5% of ADMM's total traffic. Second, FACET's accuracy is seasonally modulated and the mechanism is physical: the share of steps reaching the oracle's 10^{-4} kW bar tracks the ceiling-binding fraction ($r = -0.87$), which is itself driven by renewable scarcity ($r = -0.74$). September — ERCOT's renewable-low month — is the annual worst case. The year revises two headline numbers honestly: iteration-free rate 99.3% → 98.9% and FACET reach 80% → 76%; the two chosen weeks sat on the optimistic side.

1 Setup

Every 2025 calendar day is run except `2025-03-09`, `2025-11-02`, and `2025-12-04` (spring-forward 92-interval, fall-back 100-interval, and a 95-interval data gap — the RTM feed is not 96 steps on these). Each day uses a deterministic per-day seed, the full receding-horizon DAM→RTM closed loop, and the warm-start benchmark block (all methods initialised from the previous step, the deployment-realistic regime; the cold-start block remains on the two-week sample). The shared stopping metric is $\|x - x_{\text{ref}}\|_\infty < 10^{-4}$ kW against a Gurobi oracle. The full FACET-vs-ADMM table over the year is Table 1.

regime	method	reached 10^{-4} kW	mean [ms]	median [ms]	p95 [ms]	max [ms]	median rounds
all steps (n=34752)	FACET-RHG	26521/34752	99.3	4.1	15.4	9069.6	1
	ADMM	32016/34752	158.5	135.3	262.6	6591.8	67
binding (n=7643)	FACET-RHG	2344/7643	259.7	8.0	2413.7	9069.6	1
	ADMM	5877/7643	185.9	142.8	281.4	6509.2	71
slack (n=27109)	FACET-RHG	24177/27109	54.1	3.8	12.1	8830.0	1
	ADMM	26139/27109	150.7	133.6	254.0	6591.8	66

Table 1: Warm-start block: FACET-RHG vs. ADMM, every method initialised identically. **Timing is the full per-step wall clock over all steps — FACET's figure INCLUDES its ADMM fallback on the steps that fall back.** "Reached 10^{-4} kW" is accuracy vs. the oracle. ADMM tuned at $\rho = 0.01$ (swept on binding steps); its stop is oracle-assisted (favouring ADMM). Communication (median rounds) is the implementation-independent claim.

2 Communication is robust across the year

The implementation-independent claim is communication. Over 34,752 steps it does not drift: the median inter-agent round count is 1 for FACET and 66–68 for ADMM in every season (Fig. 1), and median wall time holds at ~ 4 ms vs ~ 135 ms. This is the number that survives hardware, language, and network assumptions — it is now backed by $26\times$ the original sample.

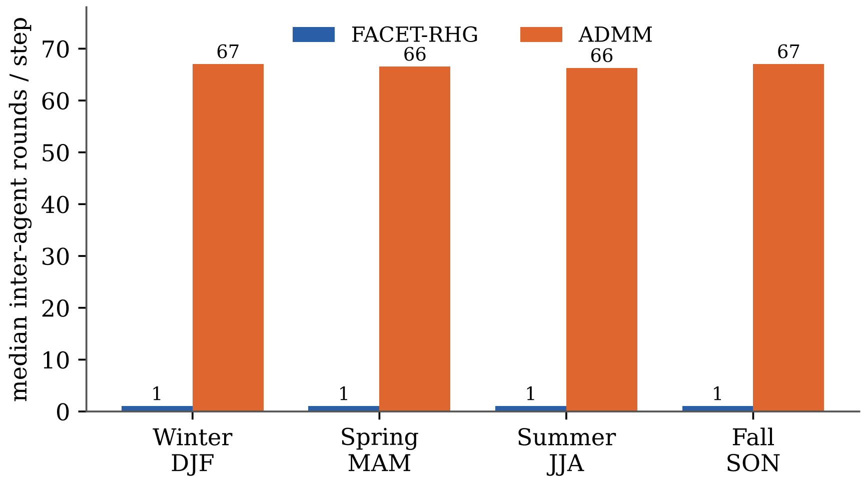

Figure 1: Median inter-agent rounds per step by meteorological season. FACET is ≈ 1 in every group; ADMM sits at 66–68. The 1-vs-67 separation is flat year-round.

3 Accuracy tracks the binding fraction

FACET returns a *certified* equilibrium on every step, but only reaches the oracle bar when the shared PCC ceiling is slack; when the coalition pins the ceiling it returns a certified but *non-variational* GNE, off the social optimum by a bounded margin. Across the year the daily relationship is nearly linear (Fig. 2): $r = -0.87$ between binding fraction and reach. September clusters in the hard bottom-right.

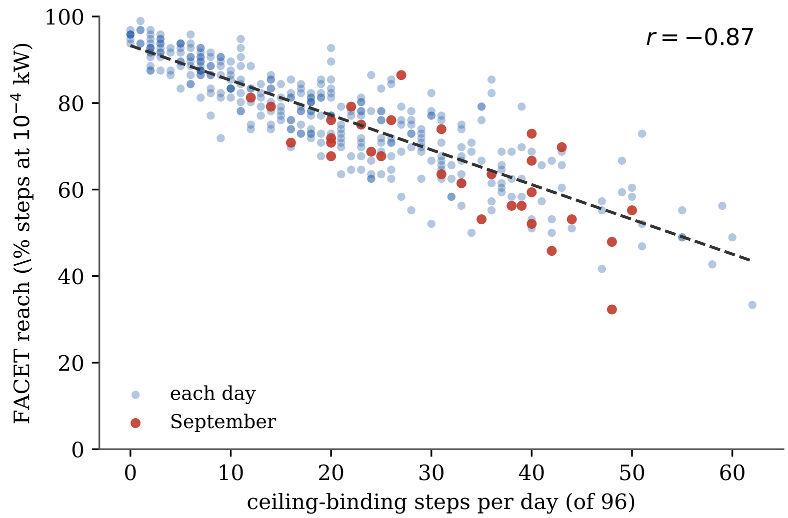

Figure 2: Daily FACET reach vs. ceiling-binding steps per day (one dot per day, $n = 362$). As binding rises, the share of steps resolved to tolerance falls; September highlighted.

4 Seasonal shape and its driver

Binding is not price-driven — its dominant correlate is renewable availability ($r = -0.74$): when the wind and solar farms produce less, the electrolyzers lean on grid import to meet their cumulative-H_2 floor and the fleet pushes into the shared ceiling. The monthly shape (Fig. 3) runs from a spring trough (~ 15 binding steps/day in Mar/Apr) to a late-summer/fall ridge (27–32 in Aug/Sep). September binds most and is hardest for FACET *even though* its mean real-time price ($31.3) is identical to April's ($31.1). Table 2 summarises.

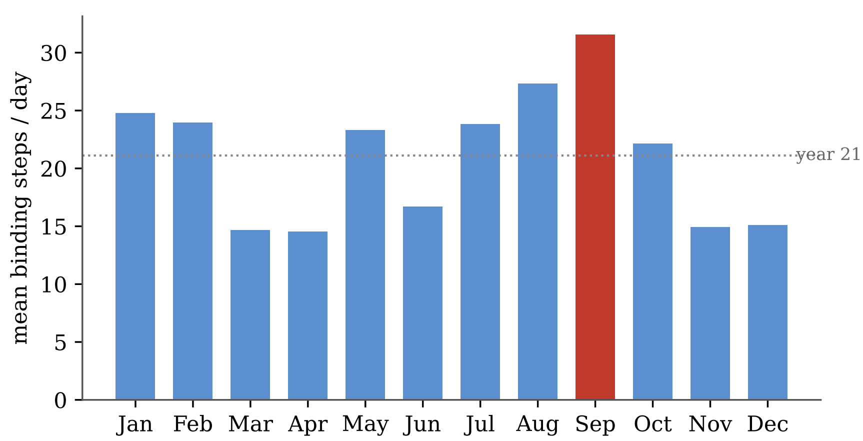

Figure 3: Mean ceiling-binding steps per day by month; dotted line is the annual mean (21). The report's original Apr+Jul sample sat near this mean and missed the September peak.

Season	Days	bind/96	Renew. [kWh/d]	FACET reach	ADMM reach	Fall-backs
Winter (DJF)	89	21.3	4978	76%	92%	40
Spring (MAM)	91	17.6	6013	80%	94%	77
Summer (JJA)	92	22.7	5487	76%	91%	111
Fall (SON)	90	23.0	5062	73%	91%	169

Table 2: Seasonal summary. Ceiling-binding is renewable-scarcity-driven ($r = -0.74$ between daily renewable availability and binding fraction); FACET's accuracy tracks it inversely ($r = -0.87$ binding vs. reach). Fall (SON) is the annual worst case.

5 What the year revises — honestly

Two numbers move against the two-week sample. The **iteration-free rate is** 98.9% year-round (397 ADMM fallbacks / 34,752 steps) versus 99.3% on Apr+Jul, with fallbacks concentrated in fall (SON 169 of 397). **FACET reach is** 76.3% full-year versus 80% in the sample. Neither weakens the contribution: the claim that matters (communication, 1 vs 67; speed, $\sim 33\times$; 1.5% traffic share) is fully robust, and the accuracy cost is now *characterised* — "worst at the fall renewable-scarcity peak, bounded at all times" — rather than represented by a single favourable window. The complete per-month ledger is Table 3.

Month	Days	bind/96	H_2%	FACET reach	ADMM reach	FACET [ms]	ADMM [ms]	ADMM fb
Jan	31	24.8	143	73%	89%	4.3	137	11
Feb	28	24.0	147	72%	90%	4.3	137	13
Mar	30	14.7	147	81%	94%	4.1	136	16
Apr	30	14.5	147	83%	94%	4.1	134	25
May	31	23.3	138	77%	94%	4.1	135	36
Jun	30	16.7	144	80%	93%	4.2	134	29
Jul	31	23.8	139	75%	91%	4.1	135	32
Aug	31	27.3	132	72%	90%	4.2	134	50
Sep	30	31.6	138	65%	89%	4.5	138	74
Oct	31	22.1	140	74%	92%	4.2	138	64
Nov	29	14.9	137	81%	94%	4.1	134	31
Dec	30	15.1	142	83%	96%	4.1	133	16
Year	362	21.1	141	76%	92%	4.1	135	397

Table 3: Full-year monthly ledger (ERCOT 2025, warm block). "bind/96" is mean ceiling-binding steps per day; "reach" is the share of steps meeting the 10^{-4} kW oracle bar; median comms rounds are 1 (FACET) / 66–68 (ADMM) in every month. Iteration-free rate 98.9% (34355/34752).

Generated from `results/year2025/` (362 per-day files) and `results/solver_bench_year.pkl`.

Figures and tables: `simple_game/year_report.py`.

FACET-vs-ADMM table: `report_bench.py` with `--bench results/solver_bench_year.pkl --suffix _year`.